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ABSTRACT

A force sensor capable of detecting both the magnitude and direction of the shear stress is proposed and demonstrated. The structure is based on an array of gallium nitride (GaN) light-emitting nanostructures and an off-the-shelf imager. The directional sensitivity originates from symmetry breaking of the nanostructures. Only a common bias is required for all GaN elements, making two-dimensional mapping of the stress relatively simple. As the stress sensing relies only on differential measurements, no stringent uniformity requirement across the sensor array is necessary.

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Tactile sensing is essential for artificial prosthetics and robotics applications.[1–5] The ability to detect both the amplitude and direction of forces when interacting with an object is critical for the perception of surface textures and delicate maneuvering actions such as gripping. Numerous approaches have been shown to be capable of discriminating the direction of the stress.[6–16] These devices typically involve the detection of the resistance or capacitance changes in response to an external force due to structural deformation or via the piezoresistive/electric effects. To differentiate the direction, each sensor pixel consists of multiple sensing elements each of which responds to forces in different directions. Recently, an optical pressure sensor based on a light-emitting piezoelectric zinc-oxide (ZnO) nanowire nanostructure was reported.[17] The light emission intensities of these nanostructures were sensitive to the external pressure, and the change can be detected using an off-the-shelf imager. Because only a common bias was needed for the entire array of light-emitting structures, a very high spatial resolution can be achieved. However, the ZnO nanowire pressure sensor cannot discriminate the direction of the force. The low sensitivity of optical detection in the ultraviolet wavelength range is also a challenge. Similar to ZnO, gallium nitride (GaN) and related alloys are also piezoelectric.[18] The emission wavelength of GaN can be tuned to the visible range to better match with off-the-shelf imagers. The ability to form heterostructures opens up new design degrees of freedom. In particular, breaking the spatial symmetry of the heterostructure in all three directions will embed the directional sensitivity directly into the material's optical properties. In this work, we propose and demonstrate a shear stress sensor based on elliptically shaped GaN nanopillar light-emitting elements to achieve this goal.[19]

Figure 1 shows the schematic of the proposed GaN shear stress sensor. The device consists of three parts: the GaN nanopillar sensing elements, the mushroom top structure for hydraulic amplification, and the image sensor for signal detection. The structure can be fabricated from a standard light-emitting diode (LED) epitaxial stack.[20] A potential fabrication sequence is to pattern the nanopillars first, form the electrical contacts (e.g., using angle deposition), planarize the sample with a sacrificial layer, deposit and pattern the mushroom top, and then remove the sacrificial layer. To bring the contact to the p-type region, a SiN insulating shell is deposited between the thin Ni/Au contact layer and the GaN nanopillar.[21] Alternatively, the GaN nanopillars can be biased optically using a vertically stacked LED at a shorter wavelength.

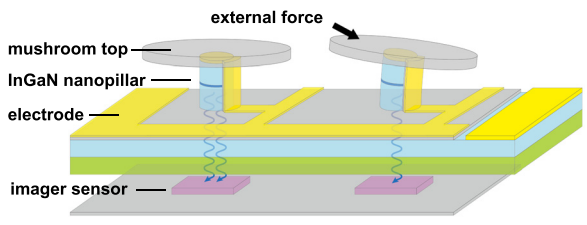


FIG. 1. The schematic of the proposed tactile shear stress sensor based on GaN nanopillar light-emitting elements.

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The operating principle of the sensor is described as follows. When the mushroom top is subject to an external force, the in-plane electric potential profile in the InGaN quantum wells (QWs) will be modified. This leads to a change in emission intensity which can be monitored by the imager. By breaking the spatial symmetry of the InGaN QWs in all three directions, the degree of the intensity change will be sensitive to the direction of the force. Using a separate imager for detection eliminates the need for any complex electrical interconnect on the GaN part. As will be shown below, only the relative change of the emission intensity of individual nanopillars, not the absolute intensity, is needed for sensing. Therefore, our design can allow a good tolerance for any nonuniformity across either the light-emitting elements or the imager.

To validate the proposed concept, we first calculated the strain distribution of the GaN nanopillar including the mushroom top and the metal/insulator layers subject to an external force using Comsol. The strain distribution in the InGaN/GaN QW region was then used to determine the electric potential profile. There are two sources of strain in the InGaN QW: the strain caused by the external stress and the residual strain caused by the lattice mismatch between InGaN and GaN.[19] Once the electric potential profile was determined in and around the QW, we solved for the Schrödinger-Poisson equation under the effective mass approximation using a commercial package nextnano$_3$. Multiple eigensolutions for the conduction and valence bands were obtained. For simplicity, we ignored the excitonic effect which is a reasonable approximation at room temperature. Finally, we calculated the matrix element to determine the emission intensity. The matrix elements from different electron and hole eigenstates were weighted by the Fermi-Dirac distribution, assuming a surface carrier concentration of $3 \times 10^{11}\,\mathrm{cm}^{-2}$ in the QW, which corresponds to a typical carrier concentration in the InGaN quantum wells in an LED. Hundreds of eigenstates were included in the calculation of the emission intensity. We assumed a temperature of 300 K throughout the simulations although it is expected that the results will be applicable at a different substrate temperature. Finally, in analyzing the results, we focused on the relative emission intensity change and ignored any wavelength shift which is difficult to detect using image sensors. We considered GaN nanopillars of different diameters and shapes (both circular and elliptical). We fixed the InGaN QW thickness to be 3 nm with 18% indium composition which will give us emission around a wavelength of 500 nm at which a typical silicon-based imager has excellent sensitivity. We assumed a SiN mushroom top with a circular geometry and a thickness of 400 nm, while the size was varied for optimizations.

Figures 2(a) and 2(b) show the results for circular GaN nanopillars of two different diameters: 150 nm and 50 nm, respectively. For simplicity, we first ignored any electrical contacts and fixed the area of the mushroom top to be 1000 times that of the nanopillar. The relative intensities were defined in reference to the intensity values with no input stress. The contour plots in Fig. 2 show the relative intensity for different magnitudes of stress in the normal and shear directions, in the range of interests (~0.1–10 kPa) for tactile sensing applications. The results show that the circular nanopillar is only sensitive to the shear but not the normal stress. The larger nanopillar structure exhibits a higher sensitivity to the shear stress. We explain the results as follows. Without external stress, the radial potential profile in the disk shaped InGaN QW confines both electrons and holes toward the center of the nanopillar.[19] In the vertical direction, the potential profile is linear and the electron and hole wavefunctions are slightly separated due to the quantum-confined Stark effect. When the external stress is in the normal direction, the potential profile in the vertical direction remains virtually unchanged because the magnitude of the residual strain ($\sim 10^{-3}$–10^{-4}) in the QW is much greater than that due to the external stress. As a result, the emission intensity remains virtually the same as shown in the inset of Fig. 2. When the external stress is in the horizontal direction, the radial potential profile in the InGaN QW is changed due to the broken symmetry. As a result, the electron and hole wavefunctions are separated in the horizontal direction, lowering the emission intensity due to a reduced overlap integral. A larger nanopillar has a weaker confinement potential for the electrons and holes in the horizontal direction and allows the electron-hole wavefunction overlap to change more significantly due to the external shear stress. Therefore, a larger nanopillar shows a more significant change for its emission intensity.

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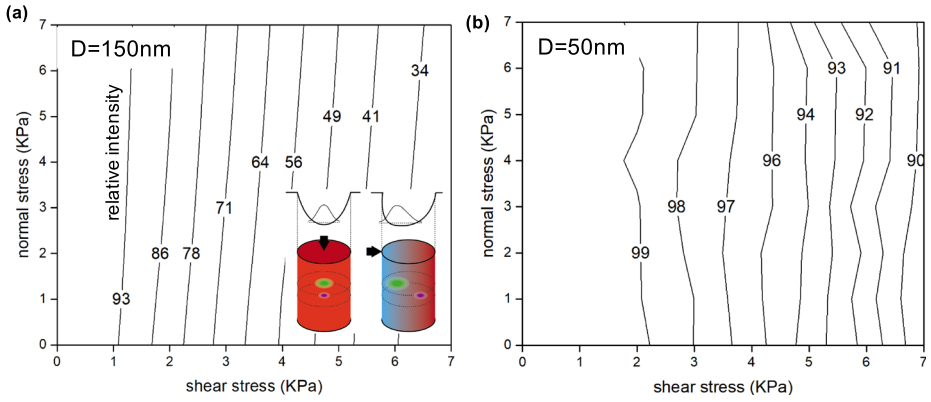


FIG. 2. The relative intensity from a circular GaN nanopillar with InGaN QW under external stress. The normal and shear directions are the directions normal and parallel to the mushroom top, respectively. (a) and (b) show the results from 150 nm to 50 nm diameter nanopillars, respectively. The inset shows the distributions of the electron (green color) and hole (purple color) wavefunctions at their lowest eigenstates under normal stress (left) and shear stress (right), respectively. The electron wavefunction (green color) is concentrated toward the top of the QW. The potential profile along the radial direction in the QW is also shown together with the electron wavefunction.

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Motivated by the results in Fig. 2, we designed a tensorial shear stress sensor using an elliptically shaped nanopillar. The potential profile in an elliptical nanopillar is asymmetric under no external stress.[20] The electrons and holes are confined more strongly along the short axis direction. As a result, the change of the emission intensity under an external stress is expected to be more significant if the stress is along the long axis direction. Figure 3 shows the relative emission intensity of a 50 nm × 200 nm elliptical nanopillar in response to a shear external stress in different directions. Here, we included the electrical contacts in the simulations. We fixed the mushroom top to have a diameter of 3 μm and centered it around the underlying nanopillar although the potential impact of misalignments to the results has also been assessed and shown to be minor (see the supplementary material). The emission intensity is only sensitive to the shear stress along the long axis and only for the direction opposite to where the electrical contact is located. This is because the movements of electrons are impeded toward the direction of the contact. Using only two elliptical nanopillar arrays with orthogonal orientations, one can deduce an arbitrary stress direction.

To demonstrate the feasibility of the proposed sensor, we performed a proof-of-concept experiment. The device consisted of an array of elliptically shaped GaN nanopillars with dimensions of 63 nm × 165 nm, height of 250 nm, and an array density of 2.7×10^9 cm^{-2}. On top of each nanopillar is a "mushroom top" structure made of chromium with dimensions of 95 nm × 230 nm, corresponding to a hydraulic amplification factor of ~2. The sample was fabricated on a standard GaN blue-emitting LED epitaxial stack on (111) Silicon grown by metal-organic chemical vapor deposition. The emission wavelength centered around 467 nm. The detailed processes of fabricating elliptical GaN nanopillars are described in Ref. 22. Figure 4(a) shows the scanning electron microscopy (SEM) image of the nanopillar array. The area around the nanopillar array is recessed to allow the strain to be generated only in the array in our measurements. The stress is applied using a double-side polished sapphire substrate coated with photoresist (SPR 220; 2.2 μm thickness) which functions as an elastomer as illustrated in Fig. 4(b). The stress is generated by moving

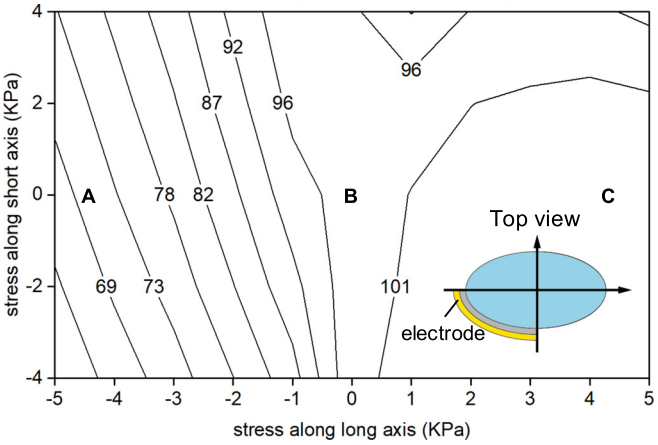


FIG. 3. The relative intensity from an elliptical GaN nanopillar under shear external stress. The position of the electrical contact is shown in the inset for a top view of the nanopillar.

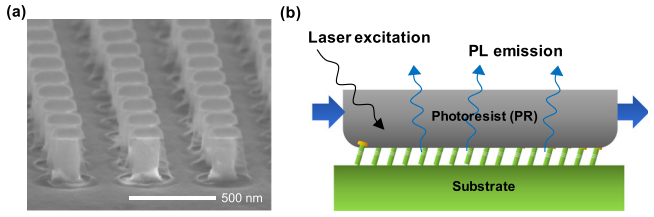


FIG. 4. (a) The scanning electron micrograph of the InGaN/GaN elliptical nanopillar array used for the proof-of-concept demonstration. The mushroom top structure is chromium. (b) The illustration of the experimental setup. The stress applicator made of a photoresist-coated sapphire substrate is mounted on a 3-axis translation stage and slided across the nanopillar array to apply stress.

the sapphire substrate with a 3-axis translation stage. When the photoresist layer is in contact with the nanopillar array, the movement of the translation stage generates stress in the photoresist layer which is then transferred to the nanopillar array. This is similar to the way a tactile sensor will likely be used. For simplicity, we biased the nanopillars optically using a 405 nm wavelength laser diode. The photoluminescence (PL) from the nanopillar array is collected by a spectrometer through the sapphire substrate and an optical fiber.

We first compared the PL spectra with and without the photoresist layer to verify the PL was indeed from the nanopillar array instead of from the photoresist. Next, we moved the translation stage using three micrometers and recorded the positions on the micrometers. We first moved the stress applicator toward the sample until we saw a change in the PL spectrum. We define this position as the origin at which the PL intensity will be compared to. We then varied the other two directions and recorded the PL spectra. We moved one translation stage at a time and reset it back to the origin as defined above before moving the next translation stage. Figure 5(a) shows the relative PL intensity (calculated by integrating the area of each PL spectrum) at a fixed vertical position. It can be observed that the PL intensity only changes significantly when the stress is along the long axis of the ellipse which agrees with the theoretical results in Fig. 3. Because the sample has no electrical contact layer, the results should only be compared to the left-hand side of Fig. 3. The measurement was repeated

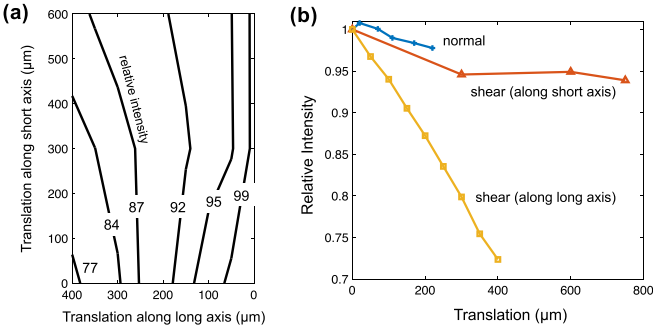


FIG. 5. (a) The relative PL intensity from the nanopillar array subject to different amounts of shear stress at a fixed vertical position. (b) The comparison of the relative PL intensity from the nanopillar array subject to different amounts of stress along the three directions. Each curve was measured with the translation stage in the other two directions fixed at 0.

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several times using different parts of the photoresist. Similar results to Fig. 5(b) that confirmed the directional sensitivity were observed. At the end of the measurements, the sample was checked by SEM again and no sign of broken nanopillars was observed.

Figure 5(b) compares the relative PL intensity change when the translation stage is moved only in one direction. It can be seen that the PL intensity from the nanopillar is most sensitive to the stress applied along the long axis of the ellipse, confirming the functionality of the proposed sensor of detecting the shear stress of both its magnitude and direction. In Fig. 5, the data are presented by the amount of translation as the absolute value of force applied on the nanopillar cannot be directly determined from the current experimental setup.

In summary, we have proposed and validated the design of an optical tactile shear stress sensor. By patterning a GaN LED chip into an array of nanopillars, the electron-hole wavefunction overlap in the InGaN QW plane can be changed according to the external stress, which in turn changes the emission intensity. As a result, monitoring the emission intensity using an imager can map out the stress distribution. Our sensor design is only sensitive to the shear stress. Moreover, using elliptically shaped nanopillars, the direction of the stress can be identified. Our design requires only a common optical or electrical bias across the entire array. All electric interconnects and circuits needed for a 2D mapping of the stress lie within the imager. The alignment with the silicon imager is not critical. By placing the imager directly underneath the GaN chip with the substrate removed minimizes beam divergence and allows a high sensor spatial resolution to be achieved. For large area applications or ones that require flexibility, e.g., in an artificial skin, an organic based imager may be a more suitable choice. As only the dynamic change of the emission intensity map is needed to detect the stress distribution, there is no need to perform a precalibration. It also enables a good tolerance to fabrication nonuniformity. Finally, we validated the proposed sensor by conducting a proof-of-concept experiment. The results agreed well with the theoretical calculations.

See the supplementary material for the detailed results of misalignment between the mushroom top and the underlying nanopillar.

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